

Indian Institute of Technology Goa

EE 635 Digital Integrated Circuit Design

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Course Project

Submitted by Group 9:

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(f) Optimize the array multiplier design such that you are able to reduce the critical path delay, or the area, or the power consumption. Demonstrate the functionality through waveforms for 2-3 different input combinations. Estimate the critical path delay through simulations. Calculate the power consumption and the transistor area for this implementation.

Ans:

Initially, the XOR gate is optimized to minimize the number of transistors, achieving an efficient design. The optimized design is shown below.

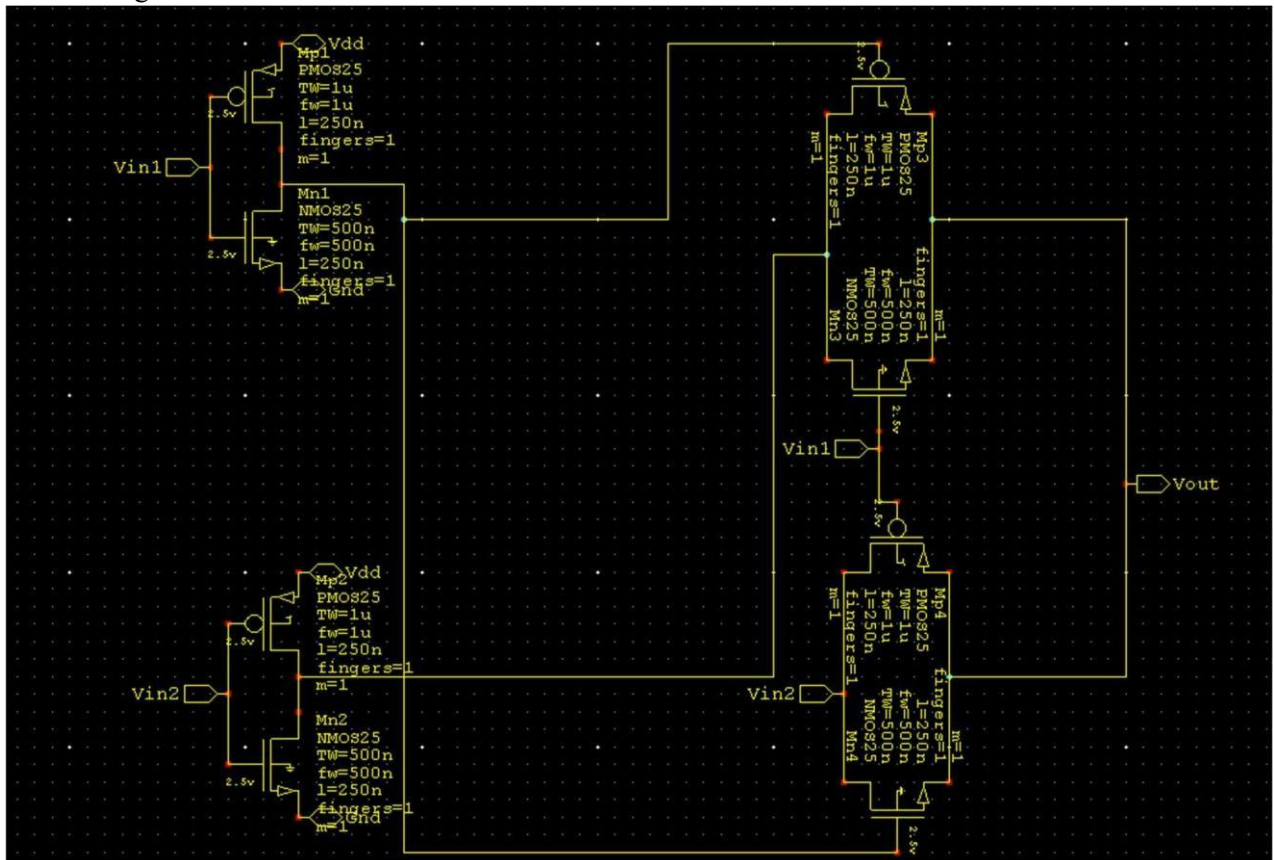


Fig-1: Optimized XOR gate

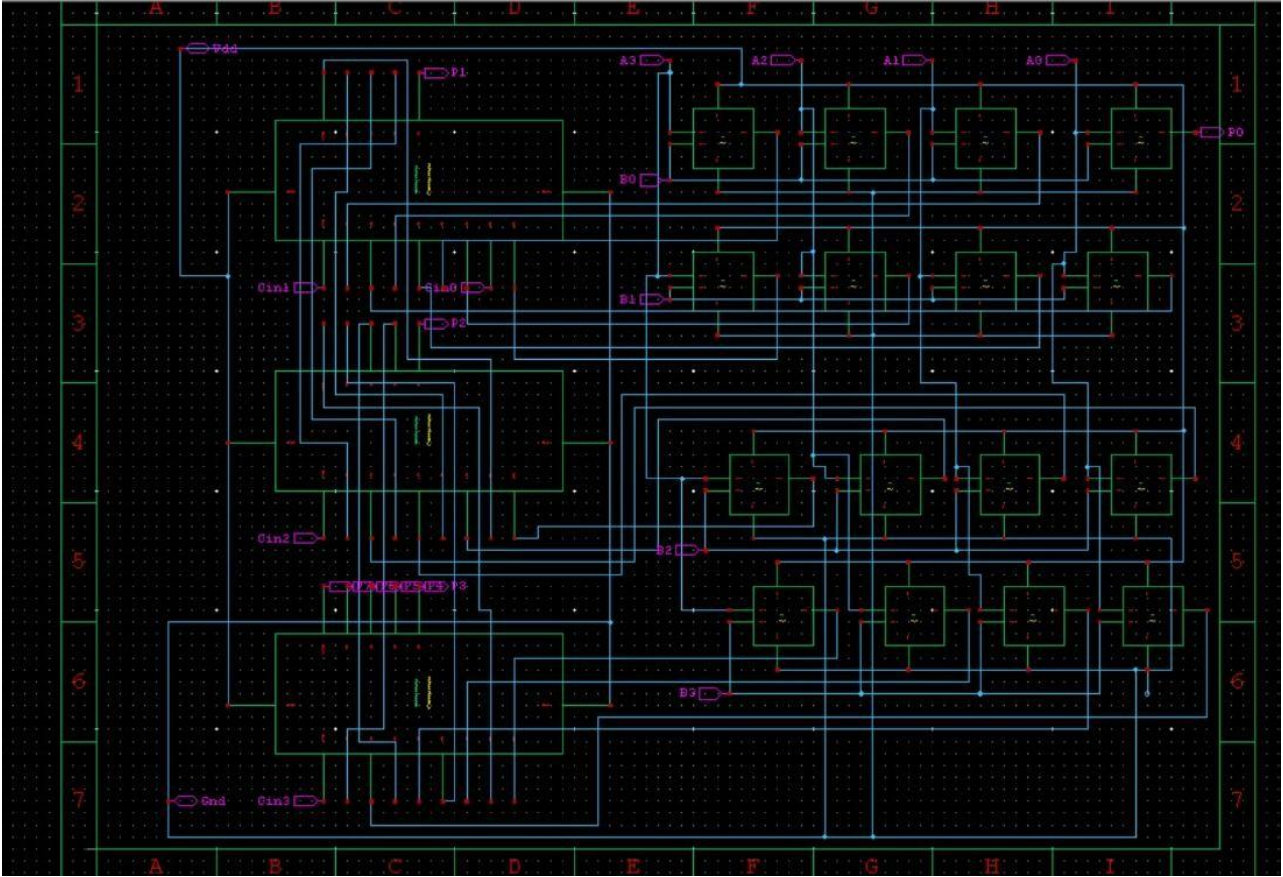
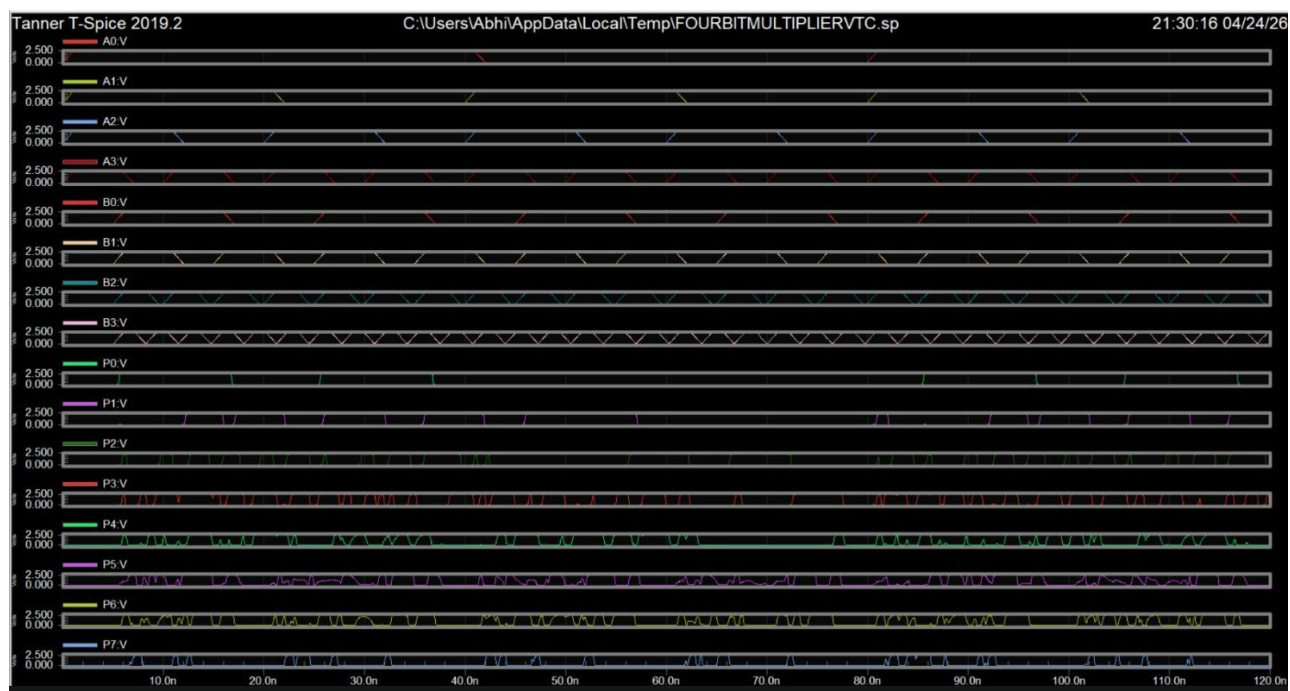
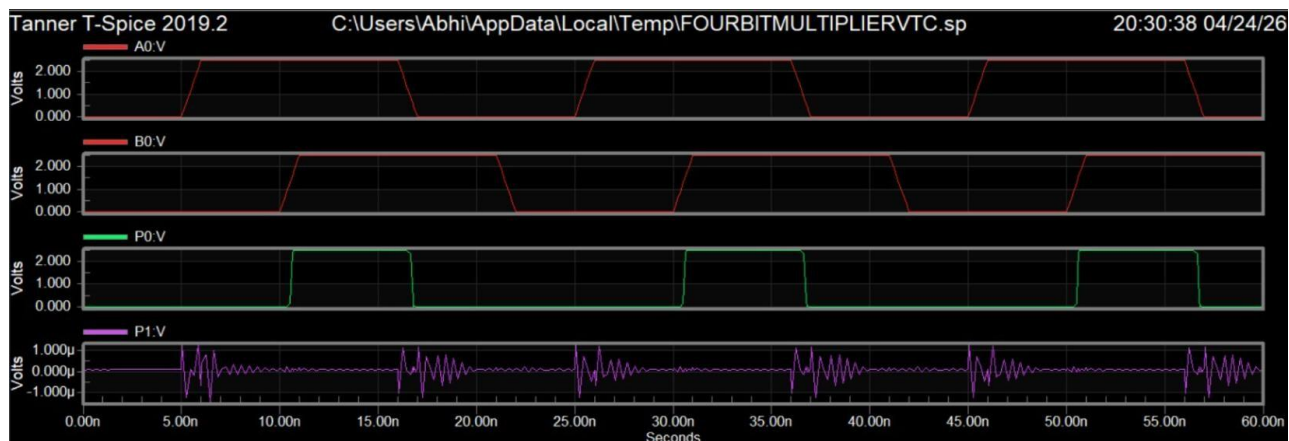


Fig-4: Four-Bit array multiplier with optimized XOR gate

In this design, the XOR gate is built using fewer transistors, which helps reduce the total number of transistors in the whole circuit. Because of this, the full adder uses only 38 transistors. When we extend this idea, a 4-bit ripple carry adder uses 152 transistors, and a 4-bit array multiplier uses 552 transistors. Using fewer transistors gives several advantages. First, the circuit becomes smaller in size, so it takes less silicon area and is cheaper to manufacture. Second, it uses less power because there are fewer components switching on and off, and the unwanted capacitance is also reduced. Overall, this makes the design more efficient and practical for VLSI systems.



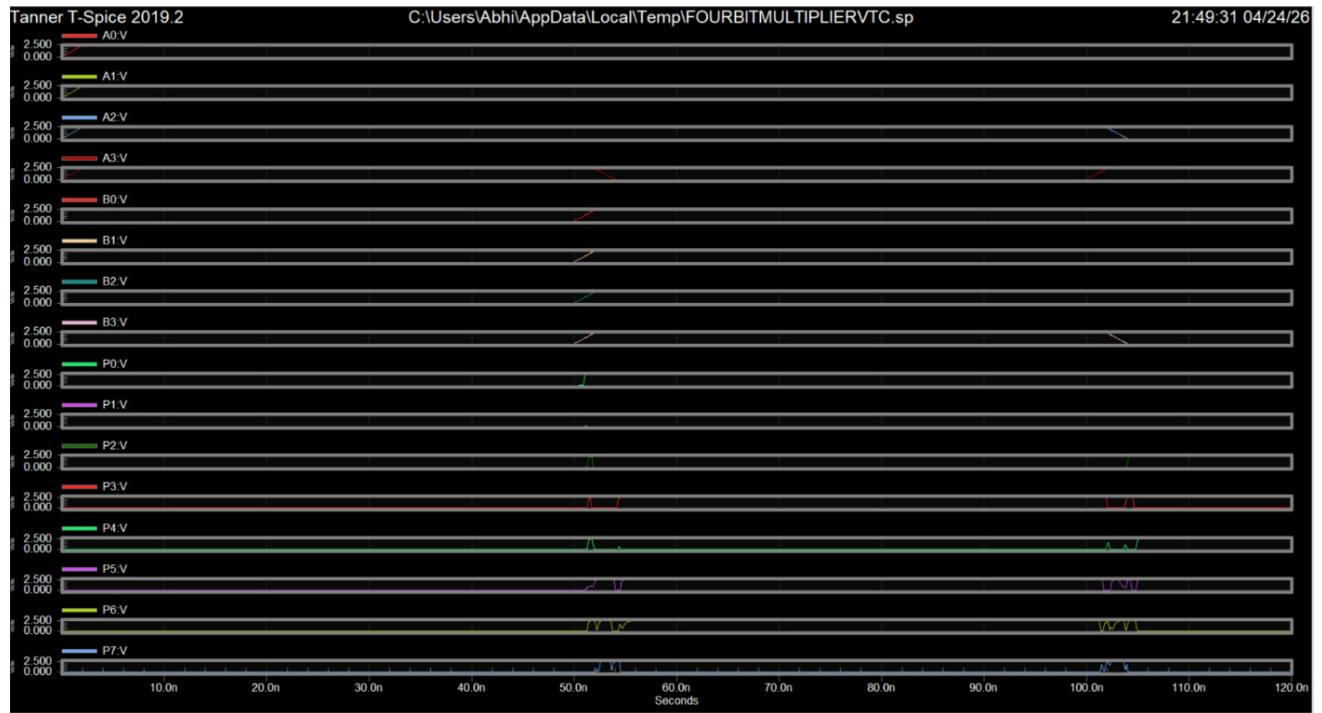


Fig-: Output waveforms at different frequency of the four-bit array multiplier .

Code used:

```
A0 A1 A2 A3 \
B0 B1 B2 B3 \
Cin0 Cin1 Cin2 Cin3 \
Gnd \
P0 P1 P2 P3 P4 P5 P6 P7 \
Vdd \
FOURBITMULTIPLIER
```

```
* ===== * POWER
```

```
* =====
```

```
Vdd Vdd Gnd DC 2.5
```

```
* =====
```

```
* INPUTS - A (SLIGHT WIDTH VARIATION)
```

```
* =====
```

```
VA0 A0 Gnd PULSE(0 2.5 0ns 1ns 1ns 120ns 240ns)
```

```
VA1 A1 Gnd PULSE(0 2.5 0ns 1ns 1ns 60ns 120ns)
```

```
VA2 A2 Gnd PULSE(0 2.5 0ns 1ns 1ns 30ns 60ns)
VA3 A3 Gnd PULSE(0 2.5 0ns 1ns 1ns 15ns 30ns)
```

```
* =====
```

```
* INPUTS - B (OFFSET + SLIGHTLY DIFFERENT WIDTH)
```

```
* =====
```

```
VB0 B0 Gnd PULSE(0 2.5 8ns 1ns 1ns 100ns 200ns)
VB1 B1 Gnd PULSE(0 2.5 8ns 1ns 1ns 50ns 100ns)
VB2 B2 Gnd PULSE(0 2.5 8ns 1ns 1ns 25ns 50ns)
VB3 B3 Gnd PULSE(0 2.5 8ns 1ns 1ns 12ns 25ns)
```

```
* ===== * CARRY-INS = 0
```

```
* =====
```

```
VC0 Cin0 Gnd DC 0
VC1 Cin1 Gnd DC 0
VC2 Cin2 Gnd DC 0
VC3 Cin3 Gnd DC 0
```

```
* ===== * ANALYSIS
```

```
* =====
```

```
.tran 0.5ns 600ns
```

```
* =====
```

```
* PLOT EVERYTHING (NOW STABLE)
```

```
* =====
```

```
.plot tran V(A0) V(A1) V(A2) V(A3)
.plot tran V(B0) V(B1) V(B2) V(B3)
.plot tran V(P0) V(P1) V(P2) V(P3) V(P4) V(P5) V(P6) V(P7)
```

```
.end clean code
```